

Monte Carlo Methods

Introduction

Generating Samples

Markov Chain Monte Carlo

Improving Efficiency

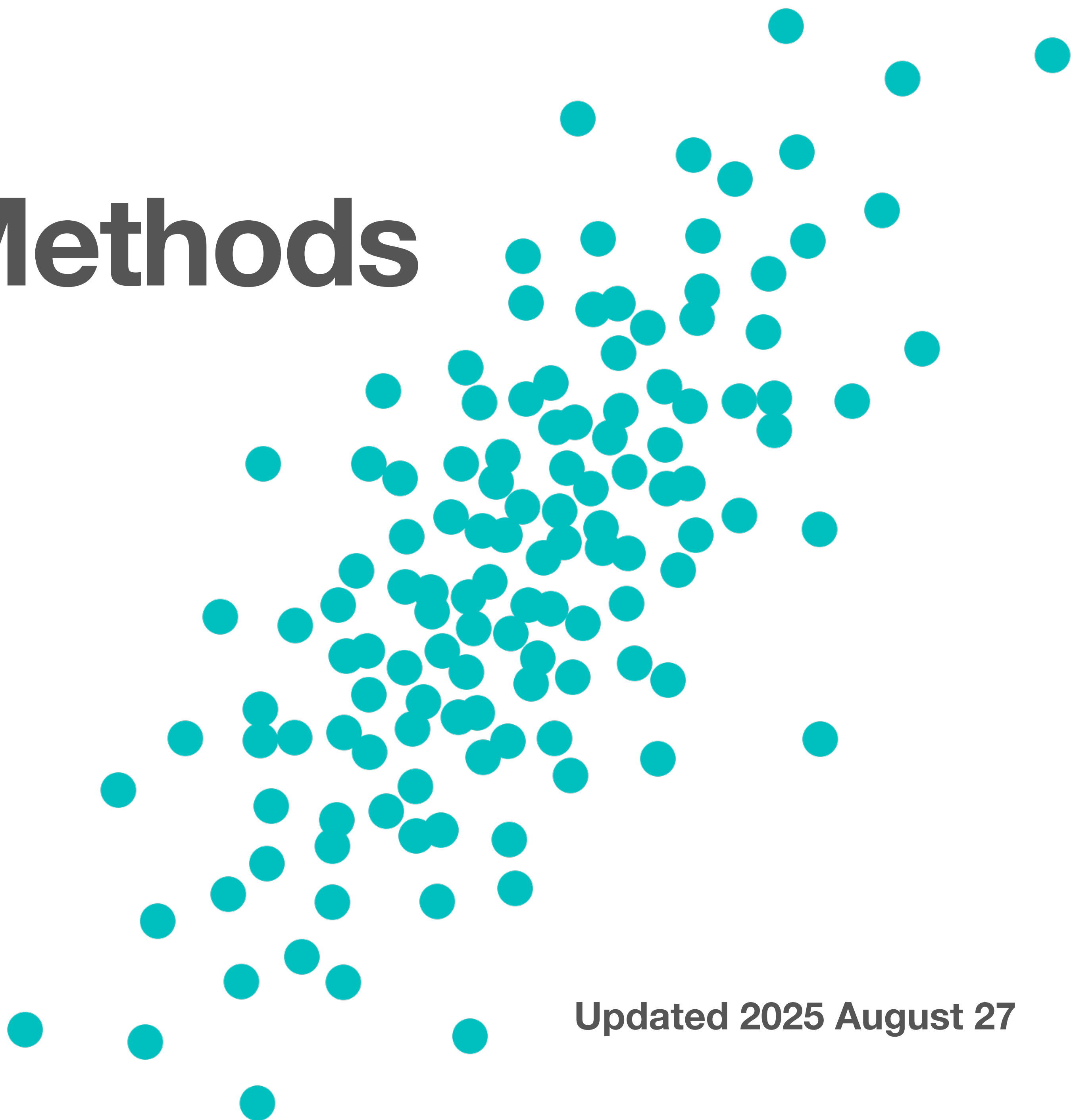
Selected Topics

Git website and repository

Canvas

Fred Hickernell, Fall 2025

Updated 2025 August 27



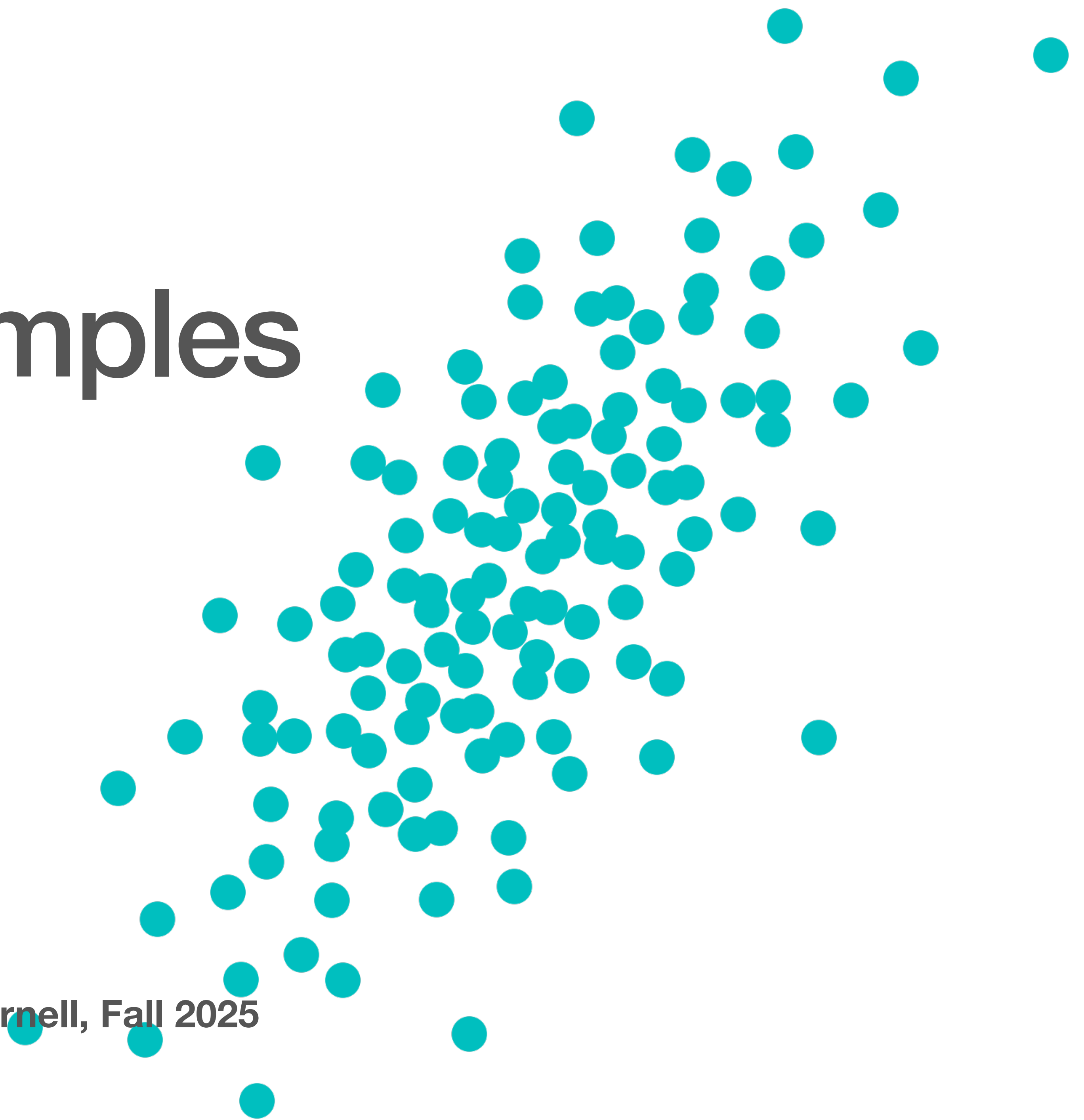
Generating Samples

Owen, Chapters 3 & 4

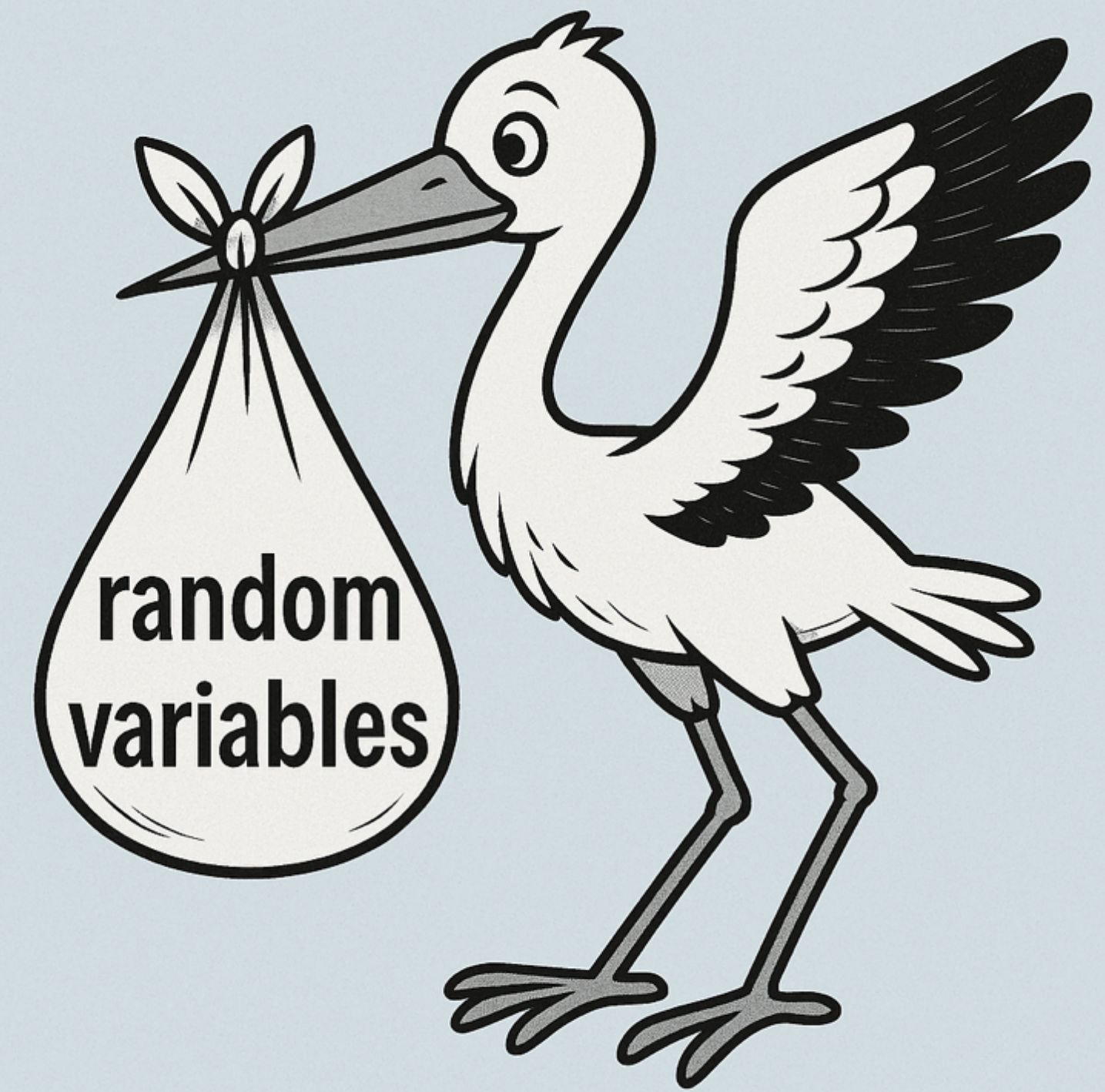
Project Selection due Oct 1

Assignment ?

MATH 565 Monte Carlo Methods, Fred Hickernell, Fall 2025



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- Algorithms generate (typically **uniform pseudo-random** numbers
- Pseudo-random number generators are tested to ensure that their output **looks IID random** in every way possible
- Care must be taken when working on **multiple processors**
- For a drastic failure see [RANDU](#)
- **Physical** random number generators may be random, but cannot be guaranteed to be IID random of the desired distribution





Quantile function gives non-uniform random numbers

- Most random number generators output $U_1, U_2, \dots$ that mimic IID $\mathcal{U}[0,1]$.
- To get $X_1, X_2, \dots$ IID with cumulative distribution function F , we use the quantile function, Q , where $Q(u) := \inf\{x \in \mathcal{X} : F(x) \geq u\}$. Note that

$$Q(u) \leq x \iff u \leq F(x)$$

- Letting $X := Q(U)$, it follows that

$$\mathbb{P}(X \leq x) = \mathbb{P}(Q(U) \leq x) = \mathbb{P}(U \leq F(x)) = F(x)$$

and so $X \sim F$, and

$$X_1 := Q(U_1), X_2 := Q(U_2), \dots \stackrel{\text{IID}}{\sim} F$$



Binomial random numbers



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x	0	1	2	3
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$x \in$	$[0, 1)$	$[1, 2)$	$[2, 3)$	$[3, \infty)$
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Note that F is **right**-continuous and Q is **left**-continuous



Explorations in generating random vectors

Generating Samples



Zero-inflated exponential

Suppose that X is a non-negative random variable with probability p_0 of being zero and otherwise an exponential distribution

$$F(x) = \begin{cases} 0 & -\infty < x < 0 \\ p_0 + (1 - p_0) \exp(-\lambda x) & 0 \leq x < \infty \end{cases}$$

To generate $X_1, X_2, \dots \stackrel{\text{IID}}{\sim} F$ from $U_1, U_2, \dots \stackrel{\text{IID}}{\sim} \mathcal{U}[0,1]$, we need the **quantile function**



Quantile function for zero-inflated exponential

$$F(x) = \begin{cases} 0 & -\infty < x < 0 \\ p_0 + (1 - p_0) \exp(-\lambda x) & 0 \leq x < \infty \end{cases}$$

$$u = F(x) \implies \begin{cases} x = 0, & 0 < u \leq p_0 \\ u = p_0 + (1 - p_0) \exp(-\lambda x), & p_0 \leq u < 1 \end{cases}$$

$$\iff \begin{cases} x = 0, & 0 < u \leq p_0 \\ \frac{u - p_0}{1 - p_0} = \exp(-\lambda x), & p_0 \leq u < 1 \end{cases}$$

$$\iff x = Q(u) = \begin{cases} 0, & 0 < u \leq p_0 \\ \lambda \log \left(\frac{1 - p_0}{u - p_0} \right), & p_0 \leq u < 1 \end{cases}$$



Random vectors with **independent** marginals

Suppose that $X = (X_1, \dots, X_d)$ is a random **vector** with independent marginals. Then

$$X = (Q_1(U_1), \dots, Q_d(U_d)), \quad U \sim \mathcal{U}[0,1]^d$$

has the desired distribution, provided that $Q_1, \dots, Q_d$ are the corresponding **quantile functions**.

We express n samples of X as an $n \times d$ matrix or array,

$$X = \begin{pmatrix} X_{11} & X_{12} & \cdots & X_{1d} \\ \vdots & \vdots & & \vdots \\ X_{n1} & X_{n2} & \cdots & X_{nd} \end{pmatrix} \text{ where } X_{ij} \text{ is the } j^{\text{th}} \text{ **component** of the } i^{\text{th}} \text{ **sample**}$$



What is your expertise in computing and probability?

- Go to menti.com
- Use code 4845 0474



Multivariate normal distribution

If $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}_d, \mathbf{I}_d)$, i.e., a d -dimensional standard normal random variable, then $\mathbf{X} = \mathbf{AZ} + \mathbf{b}$ (thinking of $\mathbf{Z}$ as a column vector) has

$$\boldsymbol{\mu} = \mathbb{E}(\mathbf{X}) = \mathbf{b}$$

$$\begin{aligned}\Sigma &= \text{cov}(\mathbf{X}) = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top] \\ &= \mathbb{E}[\mathbf{AZZ}^\top \mathbf{A}^\top] = \mathbf{AA}^\top\end{aligned}$$

So $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$, where Σ is **symmetric** and **positive-definite**. So, given a desired $\boldsymbol{\mu}$ and Σ , one needs only to find $\mathbf{A}$ with $\Sigma = \mathbf{AA}^\top$, and then

$$\mathbf{X} = \mathbf{AZ} + \boldsymbol{\mu} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$$

Note that there are **multiple** ways to decompose $\Sigma = \mathbf{AA}^\top$.



Low discrepancy sampling

If we are willing to accept samples that are not IID, then we may get substantial gains in the convergence rate. We say that

$$U_1, \dots, U_n \stackrel{\text{LD}}{\sim} \mathcal{U}[0,1]^d$$

if the **empirical distribution** of $\{U_1, \dots, U_n\}$ is close to the uniform distribution. The measure of how far away these distributions are is the **discrepancy**, and it will be defined precisely later. Some examples are

- Digital sequences, especially Sobol' sequences
- Lattice sequences
- Halton sequences
- Kronecker sequences

These come in **deterministic** and **random** forms.



Low discrepancy sampling

- LD sequences are available through [scipy](#) and [qmcipy](#), which has been developed in our research group
- Rows correspond to samples and columns to coordinates
- See [Generating Samples](#) for empirical comparisons
- The theory of LD sampling will be touched on later in the semester



Bias–variance decomposition

- Let μ be any population quantity, e.g., the mean
- Let $\hat{\mu}$ be any estimator for μ , e.g., the sample mean
- Then the mean squared error of $\hat{\mu}$ is

$$\begin{aligned}\text{mse}(\hat{\mu}) &= \mathbb{E}[(\mu - \hat{\mu})^2] = \mathbb{E}[\{\mu - \mathbb{E}(\hat{\mu})\} + \{\mathbb{E}(\hat{\mu}) - \hat{\mu}\}]^2 \\ &= \mathbb{E}[\{\mu - \mathbb{E}(\hat{\mu})\}^2] + \mathbb{E}[\{\mathbb{E}(\hat{\mu}) - \hat{\mu}\}^2] + 2\mathbb{E}[\{\mu - \mathbb{E}(\hat{\mu})\}\{\mathbb{E}(\hat{\mu}) - \hat{\mu}\}] \\ &= [\text{bias}(\hat{\mu})]^2 + \text{var}(\hat{\mu}), \quad \text{where } \text{bias}(\hat{\mu}) := \mu - \mathbb{E}(\hat{\mu})\end{aligned}$$

IID sample means have **zero bias** and positive variance

Deterministic LD sample means have positive bias but **zero variance**



Acceptance–rejection sampling